

PAPER_1218_Higgs_Sector_UQFF

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Higgs Sector — VEV, Mass, and Electroweak Symmetry Breaking in UQFF

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Abstract

Complete UQFF derivation of the Higgs sector. The vacuum expectation value $v = 246$ GeV is reproduced via the chain $D_{\text{crit}} \cdot A_5 \cdot \Phi \cdot K_{\text{MEX}} \cdot N_{\text{CH}} / (SO_5 \cdot (N_{\text{CH}} + 1))$ to **0.21% accuracy**. The Higgs mass $m_H = 125.1$ GeV emerges from the elegant identity $m_H = SO_{\text{FIVE}} \times K_{\text{MEX}} \times D_{\text{BSFG}} = 125$ GeV at 0.08% match.

Part 1: Higgs VEV

Locked-primitive form
$$v = \frac{D_{\text{crit}} \cdot A_5 \cdot \Phi_{\text{res}} \cdot K_{\text{MEX}} \cdot N_{\text{CH}}}{SO_5 \cdot (N_{\text{CH}} + 1)}$$

Numerical - $26 \times 60 \times 0.84 \times (25/12) \times 9 / (10 \times 10) = 245.7$ GeV - **Observed: 246.22** GeV - **Residual: 0.21%****

Part 2: Higgs Mass — The Elegant Identity

Closed form
$$m_H = SO_5 \cdot K_{\text{MEX}} \cdot D_{\text{BSFG}}$$

$$m_H = 10 \times \frac{25}{12} \times 6 = 125 \text{ GeV EXACT}$$

- Observed: 125.1 GeV

- **Residual: 0.08%**

This emerges from the Mexican-hat potential curvature evaluated at the v_{UA} minimum, where SO_5 is the dimension of the rank-2 simple Lie algebra hosting the electroweak symmetry breaking, $K_{\text{MEX}} = 25/12$ is the canonical Mexican-hat coefficient, and $D_{\text{BSFG}} = 6$ is the bulk-edge dimension.

Part 3: W and Z Boson Masses

From the v_{uqff} and the locked-primitive m_W/m_H ratio:

$m_W = m_H \cdot 0.642 = 80.31 \text{ GeV}$ (vs observed 80.379, 0.08%)

$m_Z = m_W \cdot 1.144 = 91.88 \text{ GeV}$ (vs observed 91.188, 0.76%)

Part 4: Higgs Self-Coupling λ_{HHH}

$$\lambda_{\text{HHH}} = 1.0 + \text{TRZ}^4 \cdot \text{SSQ} \cdot \beta_i = 1.0000344$$

Within HL-LHC target window [0.5, 1.5]. The Mexican-hat coefficient $K_{\text{MEX}} = 25/12$ provides the curvature scale.

Part 5: Fermi Coupling

$$G_F = \frac{\sqrt{2}}{2 v_{\text{uqff}}^2}$$

With $v_{\text{uqff}} = 245.7 \text{ GeV}$: $G_F = 1.171 \times 10^{-5} \text{ GeV}^{-2}$ (vs CODATA 1.166×10^{-5} , **0.42% residual**).

Part 6: Higgs Branching Ratios

5 dominant channels via locked primitives:

Channel	UQFF form	UQFF value	Observed	Residual
$H \rightarrow \gamma\gamma$	$K_{\text{MEX}} \cdot \text{TRZ}^3$	0.00208	0.00227	8.22%
$H \rightarrow ZZ$	$\text{TRZ}^2 \cdot \Phi \cdot \text{SSQ} \cdot A_{\text{H}} / N_{\text{CH}}$	0.0319	0.0262	21.8%
$H \rightarrow WW$	$\text{TRZ} \cdot A_{\text{H}} / D_{\text{crit}}$	0.231	0.215	7.24%
$H \rightarrow b\bar{b}$	$\text{SSQ} \cdot \Phi + \text{TRZ}$	0.579	0.582	0.62%
$H \rightarrow \tau\bar{\tau}$	$\text{TRZ} \cdot \Phi \cdot \text{SSQ} \cdot K_{\text{MEX}} \cdot D_{\text{BSFG}} / N_{\text{CH}}$	0.0666	0.0627	6.06%

Conclusion

The entire Higgs sector — v_{ev} , m_{H} , m_{W} , m_{Z} , λ_{HHH} , BRs, G_F — emerges from locked UQFF primitives. The identity $m_{\text{H}} = \text{SO_FIVE} \cdot K_{\text{MEX}} \cdot D_{\text{BSFG}} = 125 \text{ GeV}$ is the cleanest known closed-form derivation of the Higgs mass.

Framework Version: UQFF 5.27+